Introduction

Definitions:

Invetory: The amount of material goods held by an organization to:

1. support production
2. support activities
3. sale or customer service

Inventory management: Maintain the **optimum number** or amount of each inventroy item Uninterupted production, sales and / or customer-service levels at the **minimum cost** Inventory is the largest item in the current asset category

Two forces influencing inventory:

- Supply (Supply chain efficiency)
  - bulk purchasing (econmies of scale)
  - production continuity
  - Optimize transprt and logitics
- Demand (Meet customer Demand)
  - Buffer against uncertainty
  - Seasonal Demand
  - Market speculation
- **Too much**
  - Shortage
  - Reputation damage
  - Production downtime
  - longer lead times
- **Too little**
  - obsolete or dead stock
  - Tied up capital / holding cost
  - Storage space
- **Just right**
  - Service levels
  - Hedge against variation in supply and demand
  - Smooth production

Roles of inventory

- **Cycle** stock:
  - Meet demand during each cycle (time interval between two consecutive replenishments / production runs)
- **Safety** stock (uncertainty)
  - Hedge against demand and supply variablility; ensure service levels
- **Pipeline** stock
  - Due to replenishment lead time
- **Decoupling** stock
  - Buffer between successive stages om a production process
- **Anticipation** stock (trend & seasonality)
  - Seasonal demand; seasonal production; speculation

    - Build inventory for high demand periods in periods of low demand (leaving production capacity unchanged)
- Multiple layers of business entities
- Multiple decision makers
- Multiple objectives

E.g.
- Fridge (food for a household)
- Replenishment policies
    - Some daily, some weekly, some monthly, etc.

What factors affect inventory management:
- Demand
    - Stable
    - Time varying (cyclical)
- Supply lead time:
    - Deterministic or stochastic
    - Load dependent
- Capacity
    - Ample or limited
- Review
    - Continuous or periodic
- Excess demand
    - Backordering or lost sales
    - Item substitution
- Item & customer classes
    - Single or multiple items
    - Single or multiple client classes
- Cost structure
    - Unit cost, set-up cost, holding cost, shortage cost, penalties
- Objectives
    - Service level target (unknown shortage cost)
    - Minimize cost (known shortage cost)

Inventory policy
- How much?
- When?